

Sukhwinder Singh

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SUMMARY

Computer Science undergraduate (8.41 CGPA) with expertise in scalable backend systems and deep learning architectures. Proficient in Java, Spring Boot, microservices, and PyTorch. Solved 400+ algorithmic problems across LeetCode and GeeksforGeeks; seeking Software Development Engineer (SDE) internship roles.

EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering in Computer Science & Engineering (CGPA: 8.41)

Patiala, Punjab

2023 – 2027

Government Model Senior Secondary School (GMSSS), Sector 16

Chandigarh, India

CBSE Class XII – PCM (Percentage: 89.6%)

2021 – 2023

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript (ES6+), SQL

Backend Frameworks: Spring Boot, Spring Cloud Gateway, Spring Data JPA, Hibernate, Node.js, Express.js

Frontend Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Deep Learning & Vision: PyTorch, Torchvision, CNNs, Vision Transformers (Swin), OpenCV

Tools & Cloud: Docker, Git, GitHub, Postman, AWS (S3, EC2), Maven, Linux/Bash

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design, OS, DBMS, Computer Networks

ACADEMIC PROJECTS

CloudShare | *Spring Boot, Spring Cloud, PostgreSQL, Redis, Cloudinary, Docker, React.js* [\[GitHub\]](#) [\[Live Demo\]](#) [\[API Docs\]](#)

- Designed and deployed a microservices architecture decoupling Auth, File Ingestion, Metadata Management, and Storage, orchestrating unified request routing via **Spring Cloud Gateway**.
- Secured inter-service communication and client endpoints using stateless **JWT authentication with RBAC** and automated token validation filters.
- Slashed repeat file-lookup latency by **45%** by implementing a **cache-aside Redis pattern** for high-frequency metadata queries, significantly reducing database load.
- Engineered a resilient file ingestion pipeline integrated with **Cloudinary chunked storage**, handling multipart streaming to prevent memory overflow during concurrent uploads.

CampusRide | *Spring Boot, Spring Data JPA, PostgreSQL, Redis, Hibernate, React.js* [\[GitHub\]](#) [\[Live Demo\]](#) [\[API Docs\]](#)

- Developed a multi-tenant transit reservation platform connecting students and drivers, featuring end-to-end trip state machines, booking workflows, and an admin audit panel.
- Eliminated double-booking race conditions during peak request bursts by enforcing strict transactional consistency using **Spring @Transactional isolation** and PostgreSQL row-level locks on driver schedules.
- Sustained a **sub-120ms p95 latency** under simulated peak load by resolving Hibernate N+1 query bottlenecks through DTO projections, database indexing, and HikariCP connection pooling.
- Built and validated 30+ RESTful APIs with standardized error handling and global response wrappers, fully documented through **OpenAPI/Swagger**.

CodeE2LFDDepth | *PyTorch, Python, Swin-Transformer, CNN, Sintel/ESIM Datasets*

[\[GitHub\]](#)

- Built a hybrid deep learning model combining convolutional feature extractors with a **windowed self-attention (Swin Transformer)** backbone for event-camera light-field depth reconstruction.
- Formulated a compound loss function coupling multi-scale spatial gradient penalties with L1 edge losses, sharpening depth boundary predictions across high-speed dynamic motion sequences.
- Demonstrated a **+1.8 dB PSNR and +0.06 SSIM improvement** over baseline CNN architectures by executing automated benchmark evaluations across 50 test scenes.
- Optimized training throughput and GPU memory utilization by profiling tensor operations with PyTorch hooks and implementing vectorized data-loading pipelines.

CODING PROFILES & ACHIEVEMENTS

Algorithmic Problem Solving: Solved **400+ problems** across LeetCode, GeeksforGeeks, and Code360 focusing on Dynamic Programming, Graph Algorithms, and Tree Traversals.

Profiles: [LeetCode](#) | [Codolio](#) | [Github](#)

LEADERSHIP & RESPONSIBILITY

Academic Representative (Batch 3P1) | *TIET, Patiala*

Aug 2024 – Aug 2026

- Elected student representative for a cohort of **70+ engineering students**; coordinated course schedules, communicated departmental circulars, and resolved academic and laboratory concerns with faculty.